

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457548.7169 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.013
Transit depth (Rp/Rs)^2	3.2 +/- 0.45 [%]
Orbital Inclination (inc)	89.15 +/- 0.78
Airmass coefficient 1 (a1)	1.483 +/- 0.022
Airmass coefficient 2 (a2)	-0.305 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.48 %
Best Comparison Star	None
Optimal Aperture	3.59
Optimal Annulus	14.33
Transit Duration (day)	0.0879 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.013 vs 0.1406 (deviation 2.95σ)
Duration fit vs published	0.0879 d vs 0.0975 d (deviation 4.57σ)
Mid-time O–C	3.2 min
Depth SNR	13.8
Residual scatter	1.48 %

Light curve

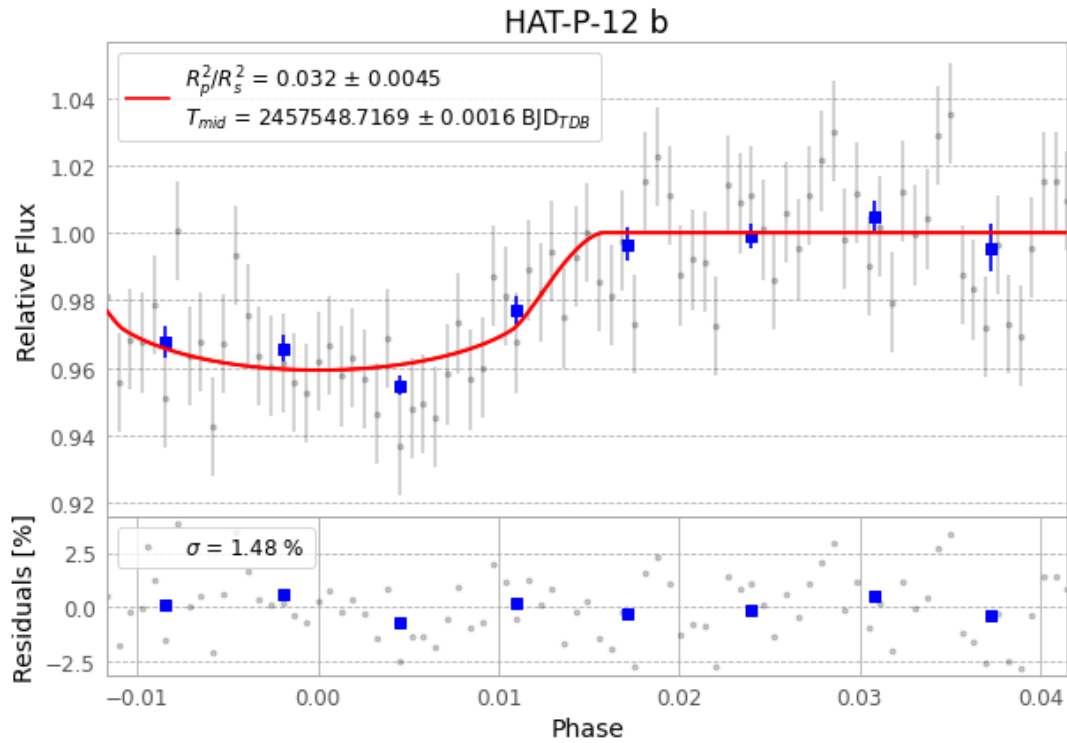


Figure 1 — FinalLightCurve_HAT-P-12 b_2016-06-08.png (SHA-256 aede4f41569efb84...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dcbf10dfe9bbb64f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), HATP-12160609041512.FITS → HATP-12160609082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-160609.log (8a27c77c30c6...), FinalLightCurve_HAT-P-12 b_2016-06-08.png (aede4f41569e...), FinalLightCurve_HAT-P-12 b_2016-06-08.csv (ba5ba42439b3...)

Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 09:36Z.